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Kaggle: Rossmann Store Sales



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November 23, 2015

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Part 1: Problem Description. Give the name of the competition you selected and write a few sentences describing the competition problem as you interpreted it. You want your writeup to be self-contained so your peer-reviewer does not need to go to Kaggle to study the competition description. Clarity is more important than detail. What's the overall goal? What does the data look like? How will the results be evaluated?

Example: "The task is to predict whether a given passenger survived the sinking of the Titanic based on various attributes including age, location of the passenger's cabin on the ship, family members, the fare they paid, and other information. Solutions are evaluated by comparing the percentage of correct answers on a test dataset."

The task is to predict daily sales 6 months in advance for 1115 Rossmann Drug Stores in Germany based on various date information such as whether it is a State Holiday (e.g Christmas, Easter) or a School Holiday and on what day of the week that date is, how many customers, types of stores (4 store models) as well as types of assortment, competitors' information such as distance and how long they have been in competition and information about promotion policy of the store.

The data provided consists of a train data, a test data including information of features for the next 6 months and a store dataset which provide descriptions of each store.

The results can be first evaluated by whether the fitting model is statistically significant and by ultimately comparing the realized sales data and the predicted sales data.

Help Center

Is the problem description clear and comprehensible?

☐ 0 pts

The submission makes no real attempt to describe the problem.

☐ 2 pts

☒ The submission makes an attempt to describe the problem, I don't really understand what is being described.

☐ 5 pts

The problem description is clear enough; I understand what is going on.

Part 2: Analysis Approach. Write a few sentences describing how you approached the problem. What techniques did you use?

Example: "I split the data by gender and handled each class separately. For the females, I trivially classified all of them as "survived." For the males, I trained a random forest as a classifier. I ignored the pclass attribute that indicated the location of the passenger's cabin because I didn't think it was relevant."

I first wanted to understand the data set. I took some descriptive statistics from each features such as whether that feature is categorical or continuous, the mean, the medium etc. Then to gain insights, I divided the 1115 stores into 223 groups of 5 stores randomly. I then plot the sales by date of all 5 stores in each group and then attach all the 5 plots in a frame graphics to see the differences and varieties of the sales evolution. I also paused 10 seconds between each frame so it took me about 40 minutes to see all the 223 frames.

Since the features are all categorical (except for the feature Customer), I colored furthermore each categories for features in each sales plot to see the variation among curves of different colors. This helped me to gain insights on which features influence the sales the most and what kind of distribution the sales might follow. I then tried to understand whether the influence has a underlying relationship with the sales or it is just a lucky correlation.

Finally, I can use a panel data estimates to fit the data to check whether these features do influence on the sales.

Is the approach to the problem described clearly? Do you have some idea how you might employ these techniques to solve the problem?

☐ 0 pts

No real attempt was made to describe the approach to the problem.

☐ 2 pts

There's a description, but I don't fully understand what it's saying.

☐ 5 pts

The description is clear enough; I understand the approach.

Part 3: Initial Solution. Write a few sentences describing how you implemented your approach. What languages and libraries did you use? What challenges did you run into?

Example: "I partitioned the data by gender manually using Excel. I used Weka to build the random forest."

I used R language for the whole analysis. I first used package "dplyr" to join the data "store" containing information about stores to the "train" and "test" data. I then transformed the "date" column as "Date" type in R by using function "as.Date()".

To gain insights from the problem, I used package "ggplot2" to generate sales plot on vertical axe and the date on horizontal axe and aesthetics color and the "geom_line" for plots. I also used "geom_smooth" to smooth the curves. I discovered the "Promo" feature influences a lot on the sales. It seemed like "Promo" captures a lot of sales variation, except for Christmas time.

For categorical features, I create dummy variables based on categories (virtually a dummy variable for each category, for every categorical features). For example, Mon variable is equal 1 if the Date is a Monday and 0 otherwise, or MonPromo variable is equal 1 if the Promo and Mon are both 1 and equal 0 otherwise.

I then created several dummy features for both "train" data and "test" data:

Mon, Tue, Wed, Thu, Fri, Sat, MonPromo, TuePromo, WedPromo, ThuPromo, FriPromo, StateHolidayA (when the state holiday is of type "a"), StateHolidayB, StateHolidayC, StoreA (when the store is of type "a"), StoreB, StoreC, AssortA (when the store's assortment is of type "a"), AssortB, AssortC. I also normalized the variable competition distance to create a new variable "ComDistNorm" and CompetitionSinceMonth and "CompetitionSinceYear" to create a new variable "ComSinceNorm". Likewise, I created new variable "Promo2SinceNorm" the normalized version of variables "Promo2SinceWeek" and "Promo2SinceYear" in the "train" data.

Finally, the regression formula looks like:

$$\log(\text{Sales}) \sim \text{Mon} + \text{Tue} + \text{Wed} + \text{Thu} + \text{Fri} + \\ \text{Promo} + \text{MonPromo} + \text{TuePromo} + \text{WedPromo} + \text{ThuPromo} + \\ \text{Promo2} + \text{Promo2SinceNorm} + \\ \text{StateHolidayA} + \text{StateHolidayB} + \text{StateHolidayC} + \\ \text{SchoolHoliday} + \\ \text{StoreB} + \text{StoreC} + \text{StoreD} + \\ \text{AssortB} + \text{AssortC} + \\ \text{ComDistNorm} + \text{ComSinceNorm}$$

Finally, I just use an ordinary linear regression to fit the data, using the "lm()" function in R with the above formula and the "train" data merged with "store" data, subset by "sales" that are positive. After that, I combine fitted coefficients and the dummy features created in the "test" data earlier to predict the daily sales for the next six months. Note that I did omit the "Customers" variable since there are no "Customers" on the "test" data. To utilize the "Customers" on the "train" data, we might try to use more complicated model.

Is the initial solution implementation clearly described? Are the tools, languages, and libraries used reasonable?

☐ 0 pts

No significant attempt was made to answer the question.

☐ 2 pts

The implementation is described, but I don't fully understand how it works.

☐ 5 pts

The implementation is well-described; I understand how it works.

Part 4: Initial Solution Analysis. Write a few sentences assessing your approach. Did it work? What do you think the problems were?

Example: "My approach did not work so well, achieving a score of 0.65. This is less than the sample solution. I suspect I should not have ignored the pclass attribute."

This approach did not work so well. Even though the chosen features are statistically significant. The R-squared value is 24.68%, which means that the fitted model only explains 24.68% of the total variation of the sales data. The reason might be because of the fact that the Ordinary Linear Regression only has an error term: the idiosyncratic error term (both in stores and times). Since there are quite a lot stores, there might be variation between stores.

Is the initial solution analysis comprehensible and reasonable?

☐ 0 pts

No significant attempt was made to assess the approach.

☐ 2 pts

There's an assessment, but I don't really understand what is being described.

☐ 5 pts

The assessment is adequately described; I understand what's going on.

Part 5: Revised Solution and Analysis. Write a few sentences describing how you improved on your solution, and whether or not it worked.

Example: "I included the pclass attribute and ignored the ticket number attribute. My score improved to 0.68."

I first tried to remove some features with Ordinary Linear Regression but the features are still significant but the R-squared is around 25% as before.

I added an error term for the set of stores and used an error component regression model for panel data with random effect. I used the "plm" function from the package "plm". In order to use "plm" function, I sorted the "train" data by "Store" ID and "Date" and then transformed the "train" data as panel data with function: "plm.data" from the package "plm". With "plm" function, the Promo2, Promo2SinceNorm and ComDistNorm and ComSinceNorm and the Intercept are not statistically significant and their variance are huge, so I tried again with "plm" without Promo2SinceNorm and ComSinceNorm. This time I have all the coefficients are statistically significant. The R-squared value is 45.22% meaning that the fitted model captures about 45.22% of the total variation. We can say that this time, the model is better fitted than the previous one. I then used the fitted coefficients to predict the daily sales 6 months in advance.

Is the revised solution and analysis presented clearly?

- ☐ 0 pts
No significant attempt was made to describe an improved solution.
- ☐ 2 pts
There's a description, but I don't fully understand it.
- ☐ 5 pts
The improvement is adequately described; I understand it.

Overall evaluation-Given the description provided, do you feel you would be able to reproduce the result?

- ☐ 0 pts
No, the description is too sketchy, incomplete, or unclear for me to be able to reproduce this result.
- ☐ 2 pts
No, but primarily because I don't have a strong enough background in the area; there is a fair amount of jargon used.
- ☐ 4 pts
Yes, I think I could reproduce the result with some effort.
- ☐ 5 pts
Yes, the solution is clearly and thoroughly described -- I could follow them easily.

Evaluating the analyses of others' is a great way to pick up new insights and viewpoints. What is one helpful thing you got out of reviewing this submission? (e.g. maybe ideas on analyzing solutions, incentive to go try this competition, etc.)?

(This question will not affect the submitter's score.) You're not expected to actually try and reproduce the submitter's result, but you may choose to do so. Did you try to reproduce the result?

- ☐ No
- ☐ Yes, but my result was different
- ☐ Yes, and my result was the same.

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